

Conformal Mappings

Sections 9.3–9.4

Complex Analysis

Review

A mapping

$$w = f(z)$$

is conformal at z_0 if

- ▶ f is analytic at z_0
- ▶ $f'(z_0) \neq 0$

Analytic functions with nonzero derivative locally behave like

$$w - f(z_0) \approx f'(z_0)(z - z_0).$$

Thus they preserve

- ▶ angles
- ▶ orientation

Constructing Conformal Maps

Complex mappings are often built by composing simple maps:

1. translations

$$w = z + a$$

2. rotations and scalings

$$w = az$$

3. powers

$$w = z^n$$

4. inversion

$$w = \frac{1}{z}$$

5. exponential and logarithm

Complex mappings of regions can often be understood step by step using these transformations.

Example: Mapping by z^2

Consider

$$w = z^2.$$

Let

$$z = re^{i\theta}.$$

Then

$$w = r^2 e^{i2\theta}.$$

Thus

- ▶ angles are doubled
- ▶ radial lines remain radial lines
- ▶ circles centered at the origin map to circles

Mapping of Regions

Suppose

$$z = x + iy.$$

Under

$$w = z^2$$

we obtain

$$w = (x^2 - y^2) + i(2xy).$$

Thus vertical and horizontal lines in the z -plane transform into curves in the w -plane.

Mapping a Strip

Consider the exponential function

$$w = e^z.$$

Let

$$z = x + iy.$$

Then

$$w = e^x(\cos y + i \sin y).$$

Thus

- ▶ horizontal lines map to circles
- ▶ vertical lines map to rays

Strip to Annulus

If

$$a < \operatorname{Re}(z) < b$$

then under

$$w = e^z$$

we obtain

$$e^a < |w| < e^b.$$

Thus the vertical strip maps to an annulus.

Mapping a Strip to a Half-Plane

Consider

$$w = e^{\pi z}.$$

If

$$0 < \operatorname{Im}(z) < 1$$

then the strip maps onto the upper half-plane.

Such transformations are often used to simplify boundary value problems.

Logarithmic Mapping

The logarithm

$$w = \log z$$

maps

- ▶ circles centered at the origin $\rightarrow$ horizontal lines
- ▶ rays from the origin $\rightarrow$ vertical lines

Thus polar coordinates become rectangular coordinates.

Mapping the Upper Half-Plane

Bilinear transformations can map the upper half-plane to the unit disk.

Example:

$$w = \frac{z - i}{z + i}.$$

Then

- ▶ the real axis maps to the unit circle
- ▶ the upper half-plane maps to the interior of the disk

Mapping the Unit Disk

Conversely,

$$z = i \frac{1 + w}{1 - w}$$

maps the unit disk to the upper half-plane.

Such transformations allow many problems to be moved between standard regions.

Strategy for Finding Conformal Maps

To construct a mapping between two regions:

1. identify boundary curves
2. choose transformations that simplify them
3. compose elementary mappings

Often the sequence is

translate $\rightarrow$ rotate $\rightarrow$ scale $\rightarrow$ apply power or exponential.

Example Strategy

Suppose we want to map a strip to the unit disk.

Possible steps:

1. map strip to half-plane using exponential
2. map half-plane to disk using Möbius transformation

The composition gives the desired conformal mapping.

Summary

Key ideas of §§9.3–9.4:

- ▶ conformal maps constructed from simple analytic functions
- ▶ common mappings include
 - ▶ powers
 - ▶ exponential
 - ▶ logarithm
 - ▶ Möbius transformations
- ▶ important region mappings:
 - ▶ strip $\rightarrow$ annulus
 - ▶ strip $\rightarrow$ half-plane
 - ▶ half-plane $\rightarrow$ disk

These mappings are central tools in solving boundary value problems.

The Riemann Mapping Theorem

One of the deepest results in complex analysis is the **Riemann Mapping Theorem**.

Theorem

Let D be a simply connected region in $\mathbb{C}$ that is not the entire complex plane.

Then there exists a conformal mapping

$$f : D \rightarrow \{ w : |w| < 1 \}$$

from D onto the unit disk.

Thus every simply connected region (except $\mathbb{C}$) is conformally equivalent to the unit disk.

Uniqueness of the Mapping

The conformal mapping in the Riemann Mapping Theorem is essentially unique.

If

$$f : D \rightarrow \mathbb{D}$$

maps D to the unit disk and

$$f(z_0) = 0.$$

The mapping is unique if we choose $f'(z_0)$ real and positive.

Conformal maps of D onto the unit disk differ only by a rotation

$$w \mapsto e^{i\theta} w.$$

Example: Rectangle to the Unit Disk

Consider the rectangle

$$0 < x < a, \quad 0 < y < b.$$

This region is simply connected, so the Riemann Mapping Theorem guarantees a conformal map to the unit disk.

The construction proceeds in stages.

1. Map the rectangle to a strip using an elliptic function.
2. Map the strip to the upper half-plane with an exponential map.
3. Map the half-plane to the unit disk using a Möbius transformation.

Thus a conformal map exists:

$$f : \text{rectangle} \rightarrow |w| < 1.$$

Remarks on the Rectangle Mapping

The explicit formula involves **Jacobi elliptic functions**.

The conformal map can be written schematically as

$$w = \frac{\operatorname{sn}(z, k) - i}{\operatorname{sn}(z, k) + i},$$

where sn is a Jacobi elliptic function.

These functions naturally arise in mappings from rectangles to half-planes.

Such mappings are widely used in

- ▶ electrostatics
- ▶ fluid flow
- ▶ heat conduction

Example: Triangle with Vertices $0, 1, i$

Consider the right triangle
with vertices

$$0, \quad 1, \quad i.$$

This region is simply connected.

Therefore the Riemann Mapping Theorem guarantees a conformal mapping

$$f : \text{triangle} \rightarrow |w| < 1.$$

Schwarz–Christoffel Mapping

The **Schwarz–Christoffel transformation** gives a conformal map from the upper half-plane onto a polygon.

If $z_1, z_2, \dots, z_n$ are points on the real axis and we want a map from the upper half-plane onto a polygon with interior angles

$$\alpha_1\pi, \alpha_2\pi, \dots, \alpha_n\pi.$$

Then a conformal map is given by

$$f(z) = A + C \int^z \prod_{k=1}^n (t - z_k)^{\alpha_k - 1} dt.$$

Interpretation of the Formula

In the Schwarz–Christoffel mapping

$$f(z) = A + C \int^z \prod_{k=1}^n (t - z_k)^{\alpha_k - 1} dt$$

- ▶ z_k are points on the real axis that map to the polygon vertices.
- ▶ $\alpha_k \pi$ is the interior angle at the k th vertex.
- ▶ A determines a translation of the polygon.
- ▶ C determines scaling and rotation.

Thus the mapping transforms

upper half-plane $\rightarrow$ polygon

with the prescribed interior angles.

Derivative Form of the Schwarz–Christoffel Mapping

The Schwarz–Christoffel transformation is often written in terms of its derivative:

$$f'(z) = C \prod_{k=1}^n (z - z_k)^{\alpha_k - 1}.$$

Here

- ▶ z_k are points on the real axis corresponding to the vertices of the polygon,
- ▶ $\alpha_k \pi$ is the interior angle at the k th vertex.

Integrating gives the mapping

$$f(z) = A + C \int^z \prod_{k=1}^n (t - z_k)^{\alpha_k - 1} dt.$$

Construction Idea

Mappings from polygonal regions are obtained using the **Schwarz–Christoffel transformation**.

For a polygon with vertices $z_1, \dots, z_n$,

$$f(z) = A + C \int \prod_{k=1}^n (z - z_k)^{\alpha_k - 1} dz$$

maps the upper half-plane to the polygon.

Here $\alpha_k \pi$ are the interior angles of the polygon.

Right Triangle Mapping

For the triangle with vertices

$$0, 1, i$$

the interior angles are

$$\frac{\pi}{2}, \quad \frac{\pi}{4}, \quad \frac{\pi}{4}.$$

Thus the Schwarz–Christoffel map has the form

$$f(z) = A + C \int z^{-1/2}(z-1)^{-3/4} dz.$$

This maps the upper half-plane onto the triangle.

Composing with a Möbius transformation gives a conformal map to the unit disk.

Summary

The Riemann Mapping Theorem states that

Every simply connected region
(except $\mathbb{C}$)
is conformally equivalent to the unit disk.

Important examples include

- ▶ rectangles
- ▶ triangles
- ▶ general polygonal regions

Explicit formulas often require

- ▶ elliptic functions
- ▶ Schwarz–Christoffel transformations